USER MANUAL

Dasti (دستی)

Academic Materials Platform

dasti.vercel.app

Version 1.0.1

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DASTI

TITLE PAGE

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Section 1: Introduction

1.1 Purpose of the Application

Dasti (دستی) is a web-based academic materials platform designed for students at FAST-NUCES, Karachi (FAST-KHI). The word "dasti" is an Urdu slang meaning something done quickly, by hand - and that is precisely the experience the platform aims to deliver.

FAST-KHI students have long maintained semester-wise GitHub repositories containing past papers, notes, assignments, quizzes, and lab materials. While these repositories are a valuable shared resource, contributing to them has historically required knowledge of Git, a GitHub account, and familiarity with forking, branching, and opening pull requests - a process that shuts out most students who simply want to share a file.

Dasti removes every one of those barriers. A student can visit the website, find their course, attach a file, and submit. The platform handles everything else automatically: creating the appropriate branch, committing the file, and opening a pull request on the relevant semester repository. The repository's maintainer then reviews and merges the contribution.

The platform also serves as a browsing interface, allowing any student to navigate semester repositories, preview files, and download materials - all without leaving the browser.

1.2 Intended Users

The primary users of Dasti are undergraduate students enrolled at FAST-NUCES, Karachi, across semesters 1 through 8. The platform is equally useful for students who want to find study materials and for students who want to contribute materials they already have. No technical knowledge is required to use the platform.

Section 2: User Profile Section

2.1 Target Audience

Dasti is built for FAST-KHI undergraduate students, primarily those in their first four years of study. This includes students in Computer Science, Software Engineering, Artificial Intelligence, and related programs enrolled in courses mapped to semesters 1 through 8 of the FAST-KHI curriculum.

2.2 Technical Background

Users are students enrolled in a Computer Science university and are expected to have general familiarity with technology and software systems. While the platform does not require any knowledge of Git, GitHub, version control, or command-line tools, users are assumed to be comfortable with common computing tasks such as using web browsers, managing files, filling out online forms, and uploading documents. Familiarity with Google Sign-In and basic web applications is expected, but no specialized technical expertise is necessary to use the platform.

2.3 User Goals

Users of Dasti typically want to accomplish one or more of the following:

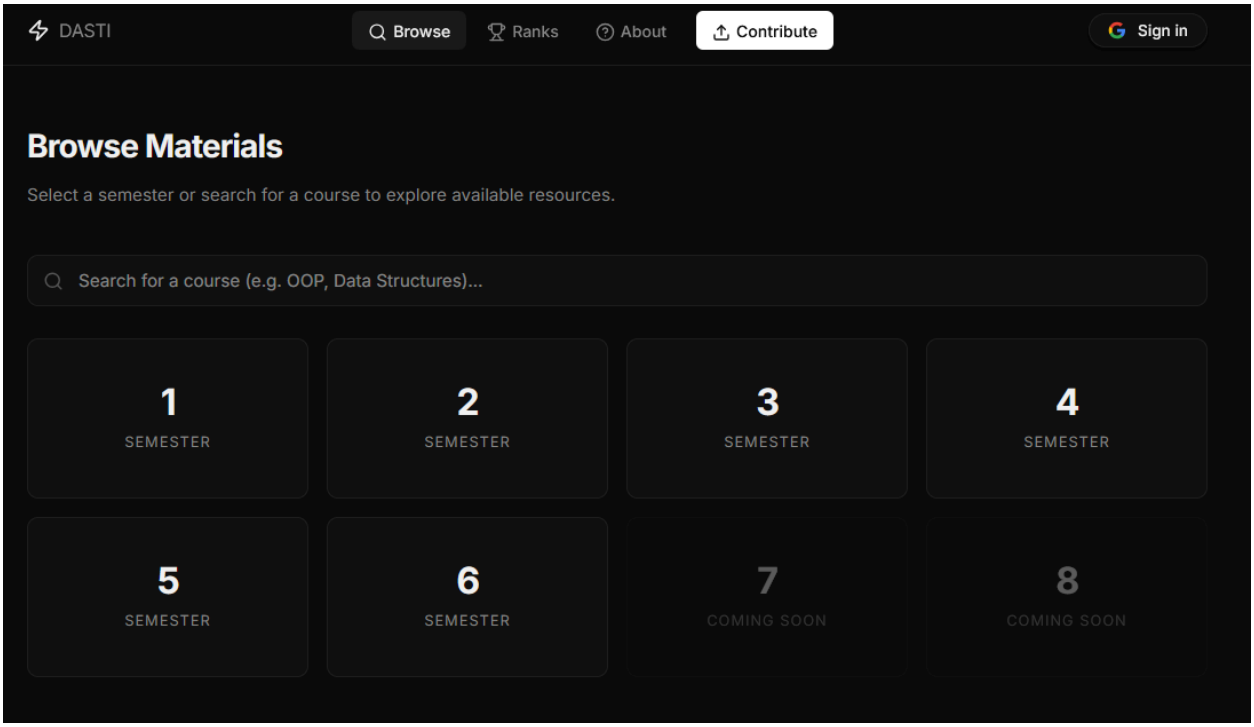
- Find past papers, notes, assignments, or other study materials for a specific course before an exam or submission deadline.
- Share materials they already have so that other students can benefit.
- Participate in the contributor community and see how their contributions rank on the leaderboard.
- Maintain a personal record of their contributions over time.

Section 3: Task-Based Instructions

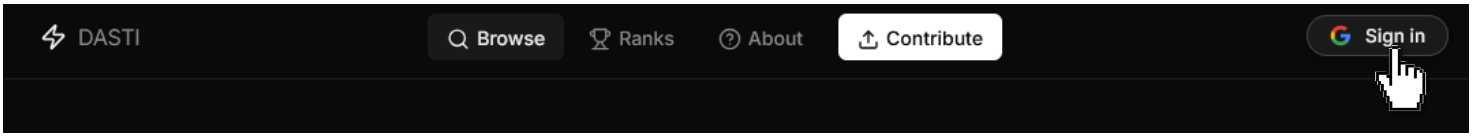
Task 1: Creating an Account (Signing In with Google)

Dasti uses Google Sign-In for authentication. There is no separate registration form - your account is created automatically the first time you sign in. You do not need a GitHub account.

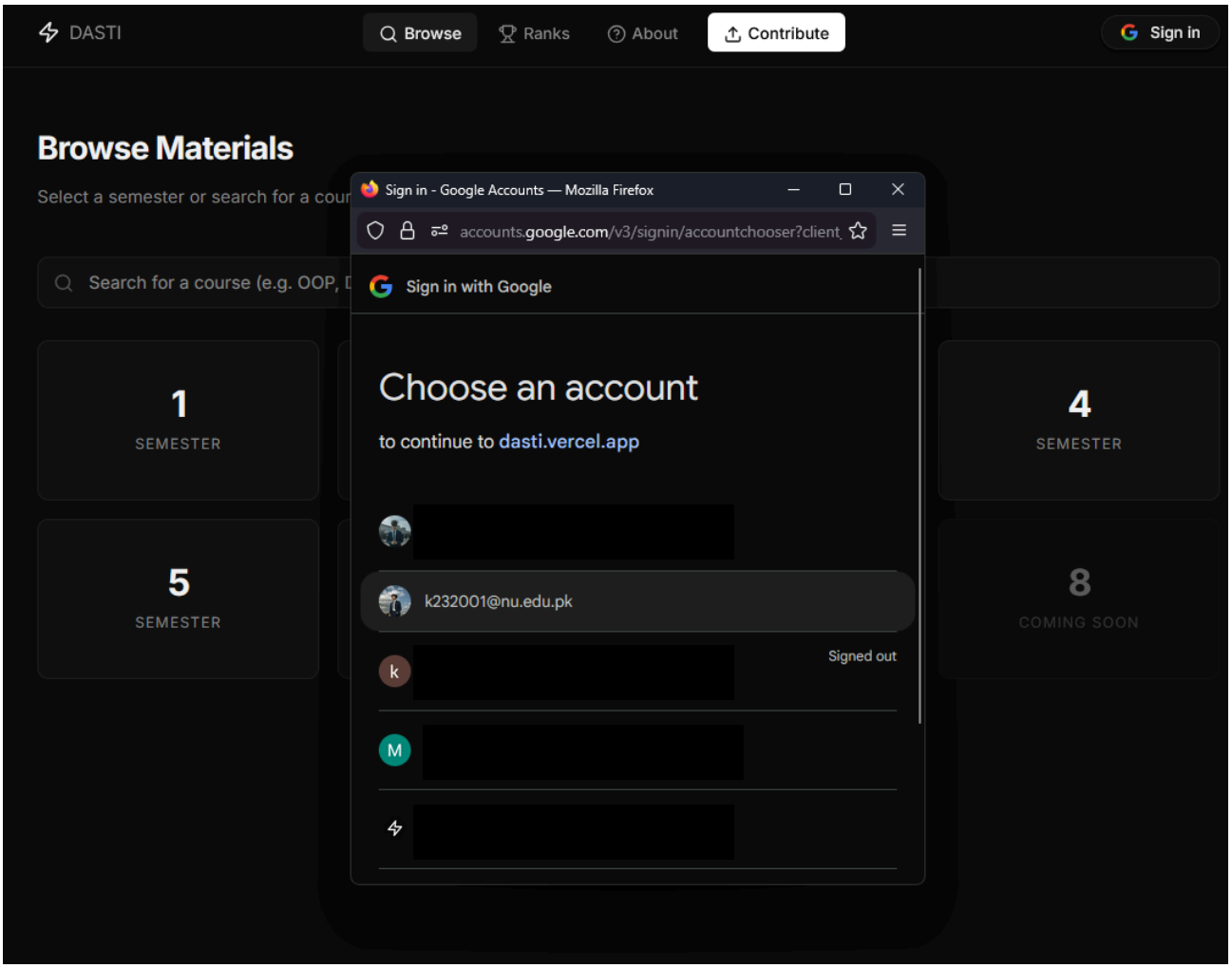
1. Open your web browser and navigate to dasti.vercel.app.



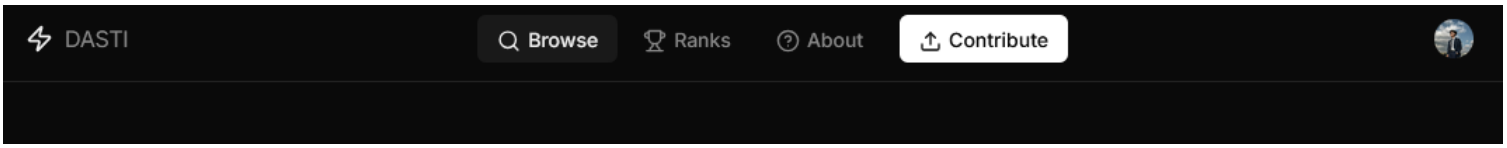
2. Locate the navigation bar at the top of the page. On the right side, find and click the Sign In button.



3. A Google Sign-In prompt will appear. Select the Google account you wish to use. If you are already signed into Google in your browser, your account will appear as a clickable option.



4. Click your account to authorize Dasti. The popup will close and you will be returned to the Dasti page, now signed in. Your name or avatar will appear in the navigation bar confirming success.



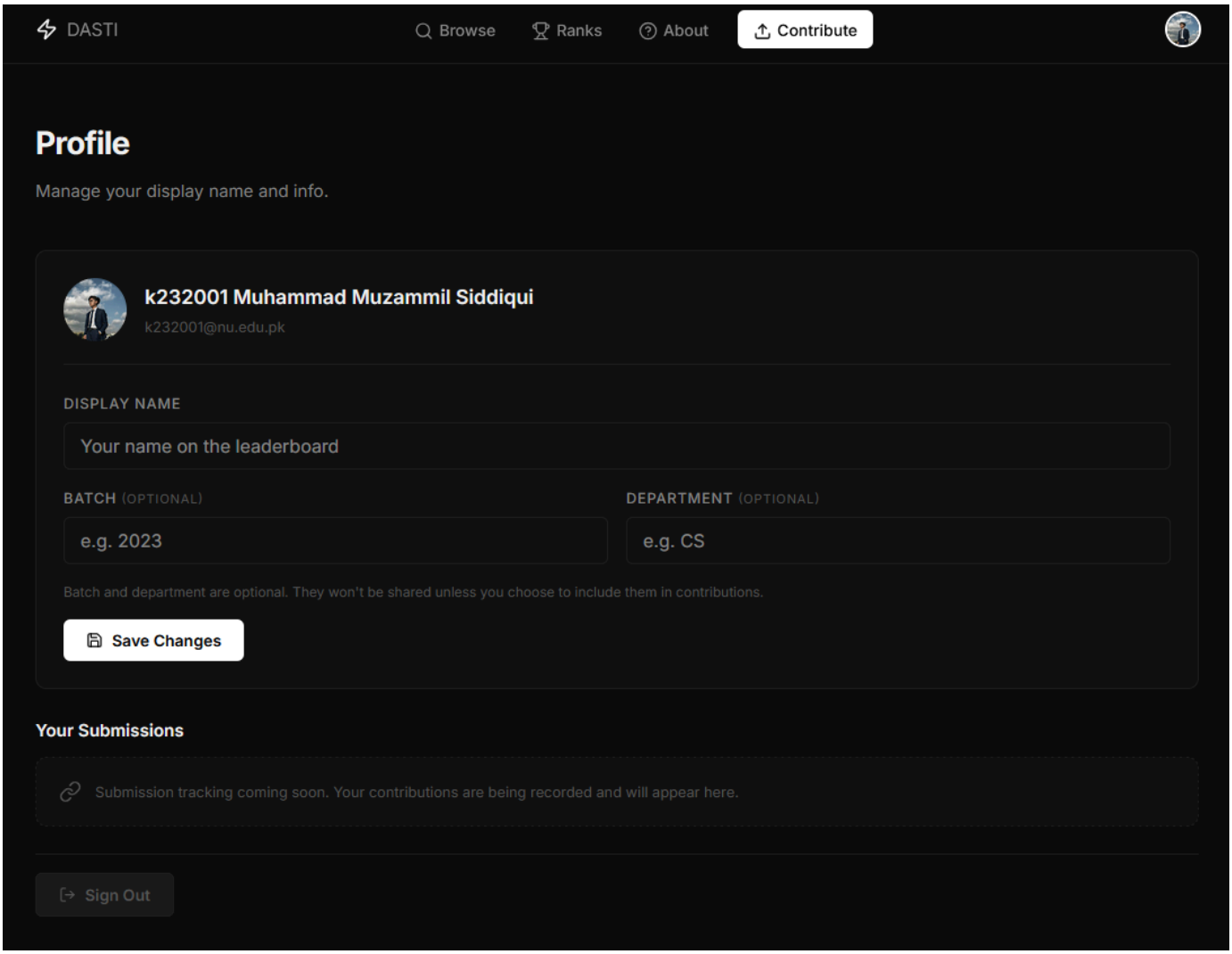
5. If this is your first time signing in, Dasti will send a welcome email to the address associated with your Google account. Check your inbox to confirm.

Note: *If you prefer not to create an account, you may still browse and download materials as a guest. However, signing in is required to contribute files.*

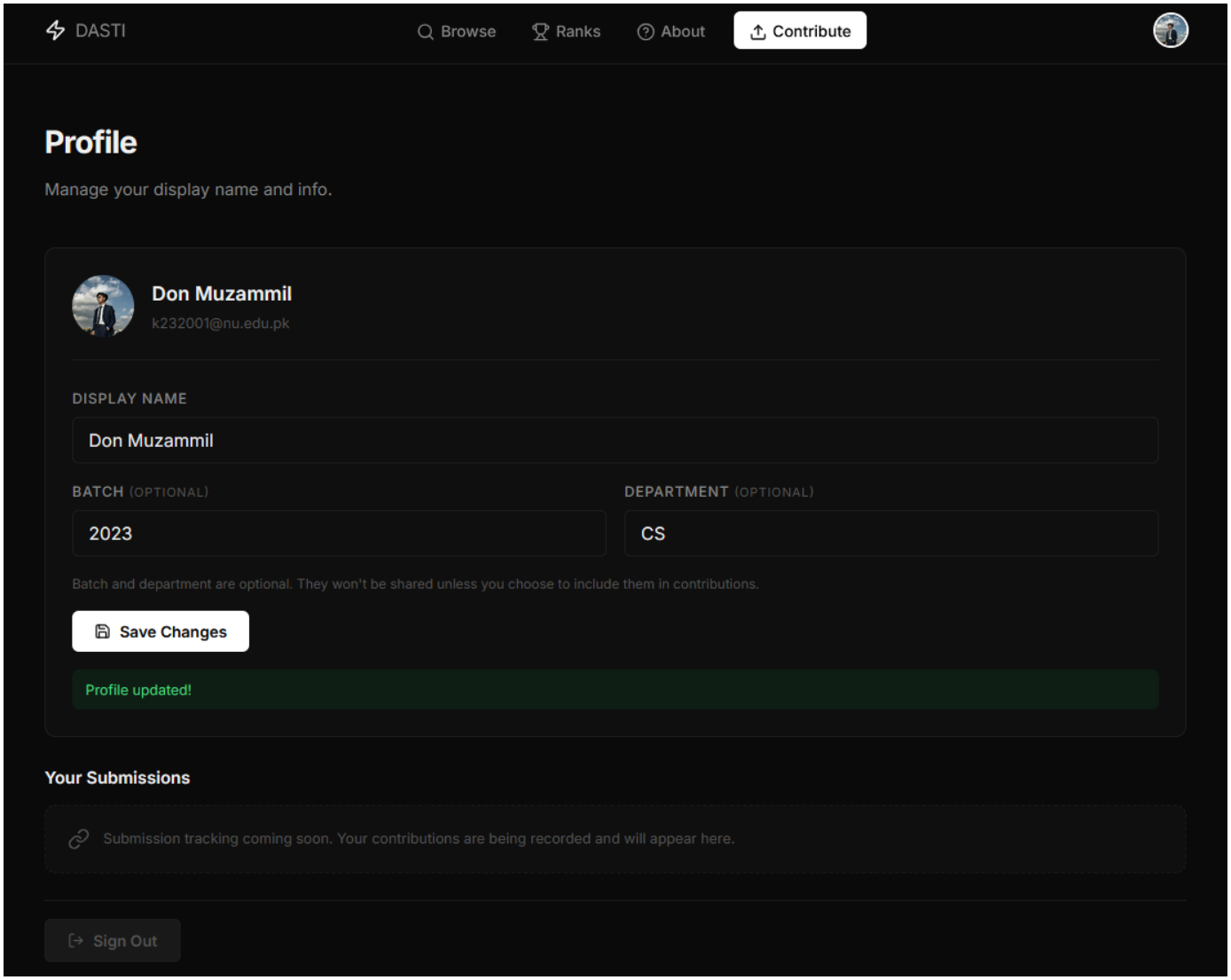
Task 2: Setting Up Your Profile

After signing in, add your batch year and department to your profile. This information appears alongside your contributions on the leaderboard.

1. Ensure you are signed in (see Task 1).
2. Click your avatar or display name in the navigation bar, or navigate directly to dasti.vercel.app/profile.



3. In the Display Name field, click and type the name you want to appear publicly.
4. In the Batch field, enter your batch year (for example, 2023). This field accepts up to 10 characters.
5. In the Department field, enter your department (for example, CS or SE). This field accepts up to 30 characters.
6. Click the Save button to apply your changes.



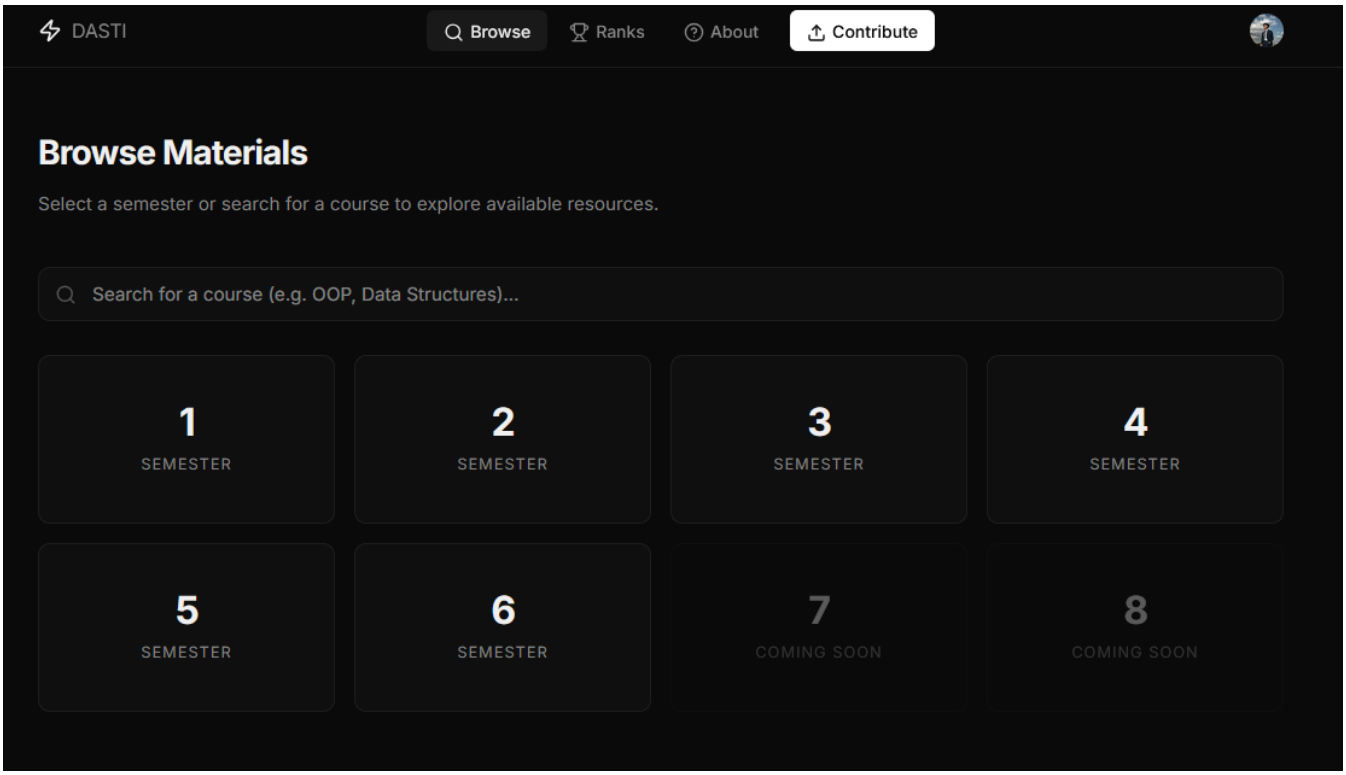
7. A confirmation message will appear indicating your profile has been updated successfully.

Note: *Your display name, batch, and department are the only editable fields. Your email is pulled from your Google account and cannot be changed here.*

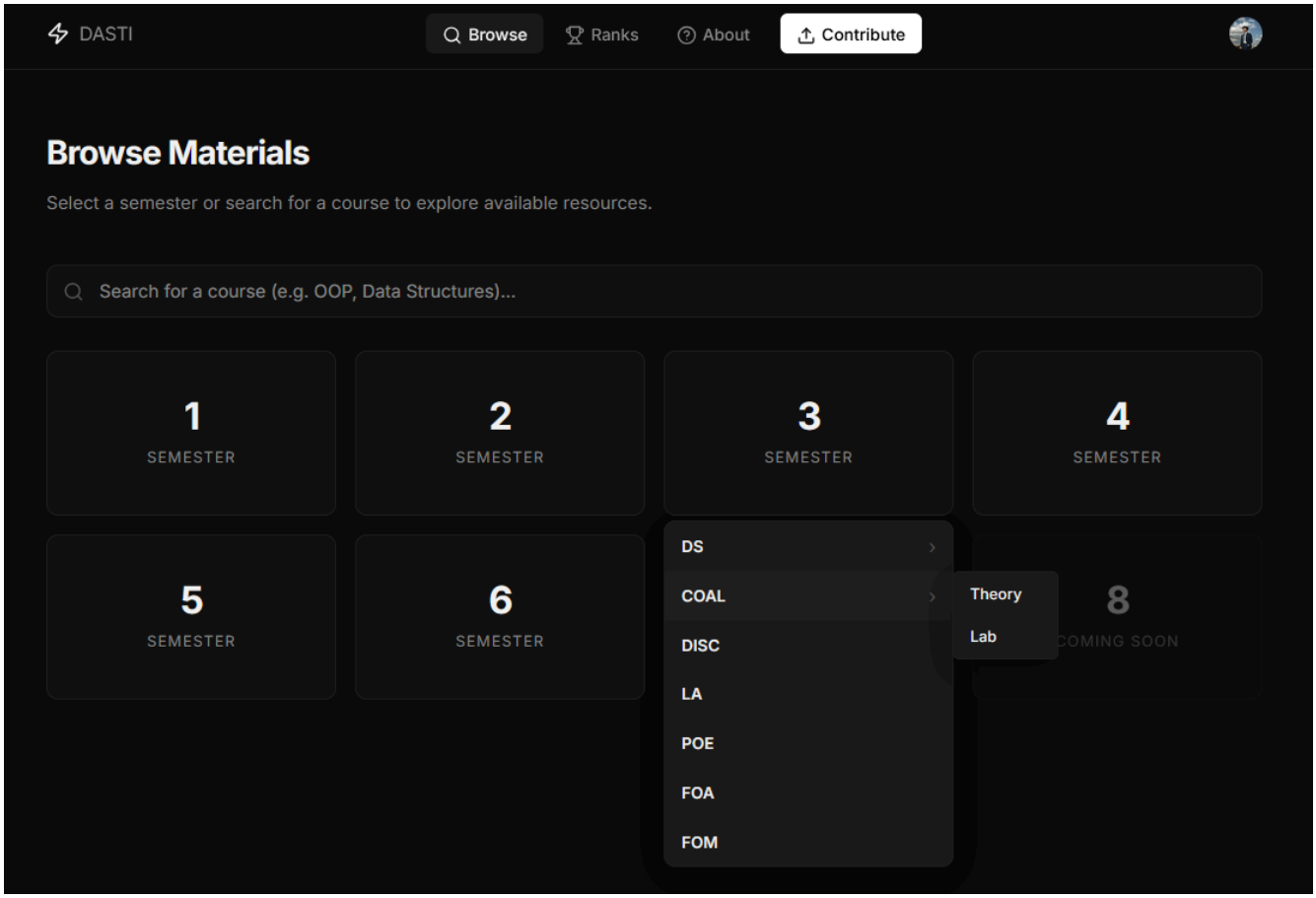
Task 3: Browsing Study Materials

Browsing lets you explore materials available in any semester repository, organized by course and material type.

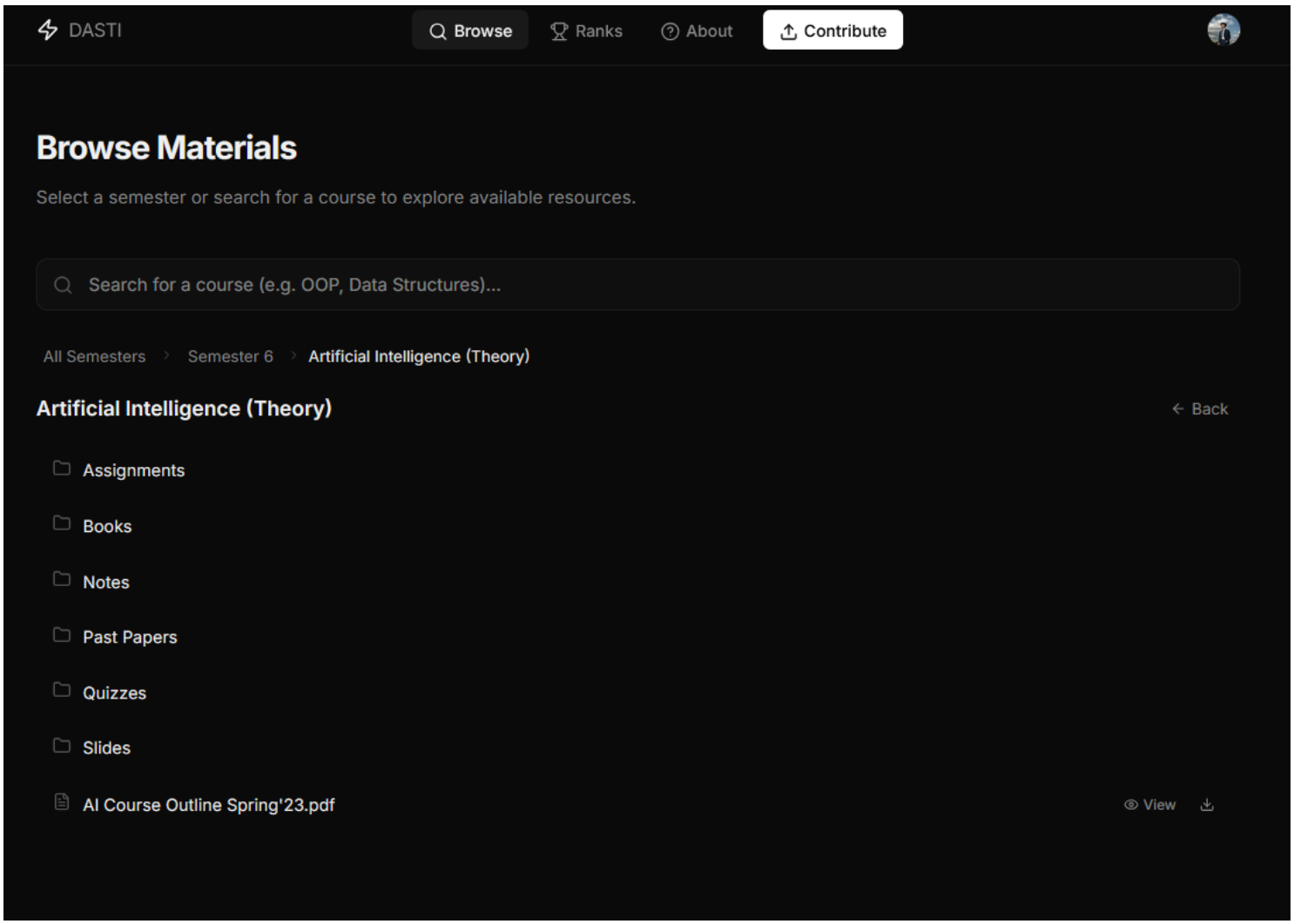
1. Navigate to dasti.vercel.app. You will see a grid of semester cards labeled Semester 1 through Semester 6, along with two placeholder cards for upcoming semesters.



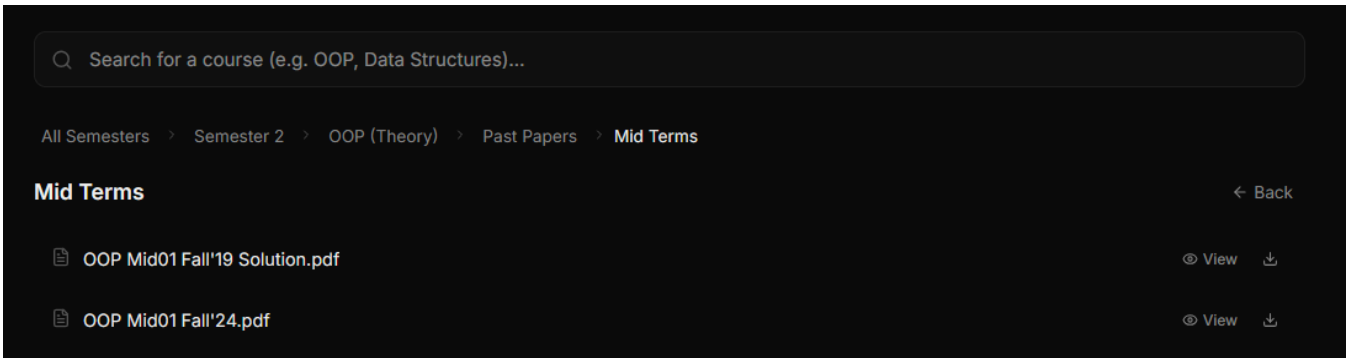
2. Hover your mouse over (on desktop) or tap (on mobile) the semester card for the semester you are interested in. A dropdown will appear listing all courses in that semester.



3. For courses that have both a Theory and a Lab folder (for example, OOP (Theory) and OOP (Lab)), a submenu will appear. Click or tap the specific option you want.
4. The page will load the contents of that course folder from the GitHub repository. You will see a list of subfolders and files sorted alphabetically, with folders appearing before individual files.



5. Click a folder name to enter it and see its contents. A breadcrumb trail at the top (for example, All Semesters > Semester 2 > OOP (Theory) > Past Papers) shows your current location. Click any segment to navigate back to that level.

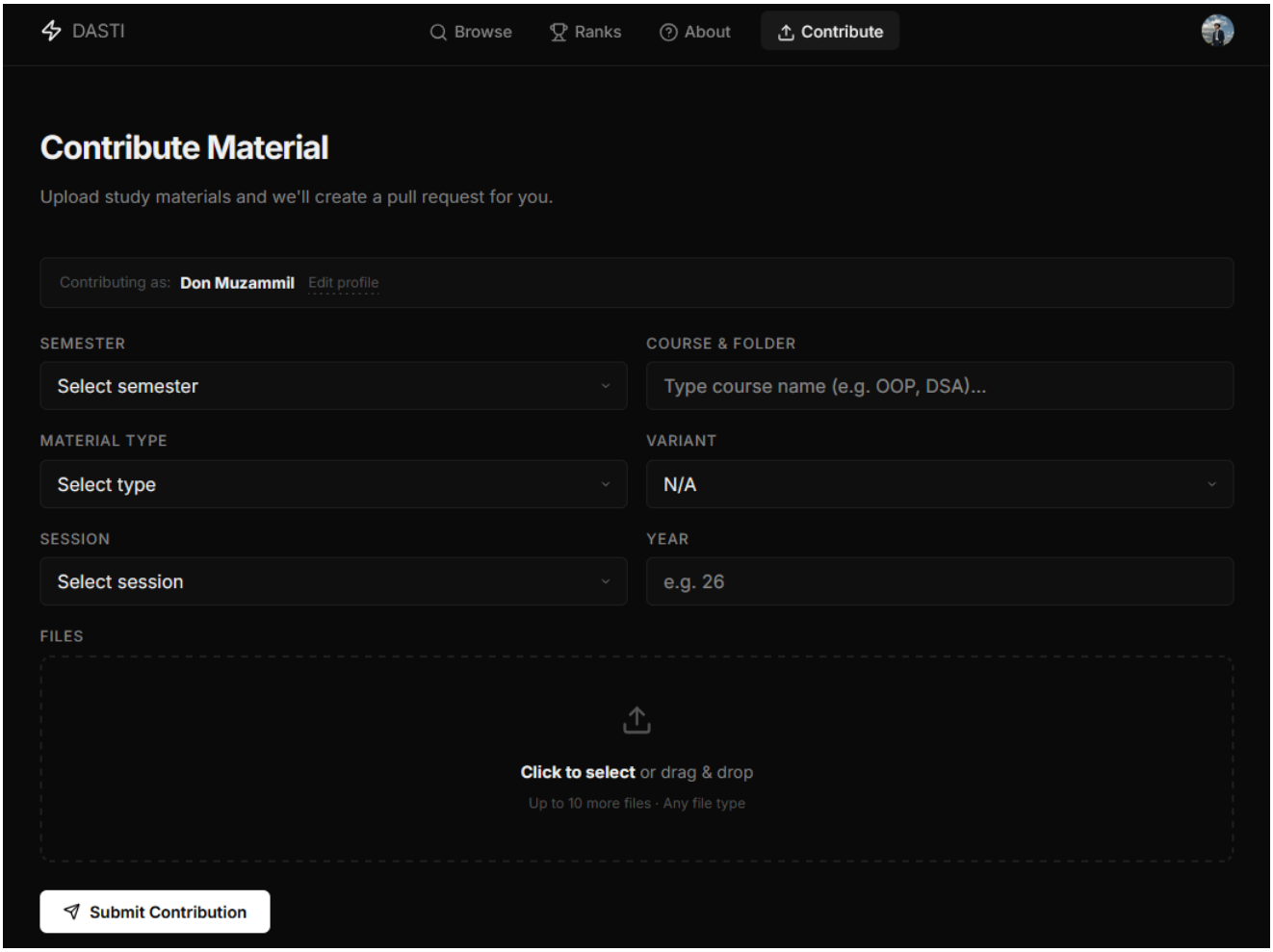


6. To return to the full semester grid, click All Semesters in the breadcrumb or click the site logo in the navbar.

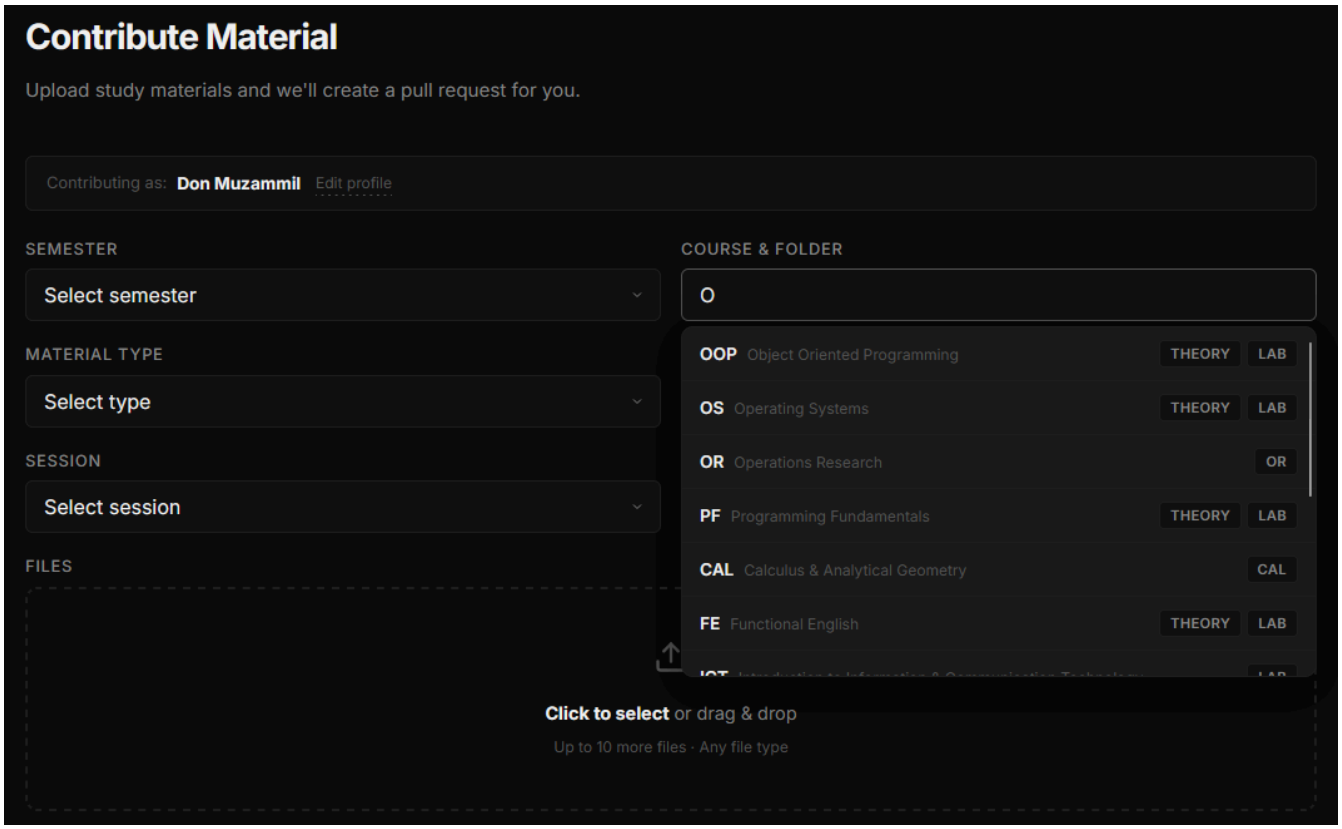
Task 4: Contributing / Uploading Study Materials

Contributing is the core function of Dasti. This task walks you through submitting one or more files for a specific course.

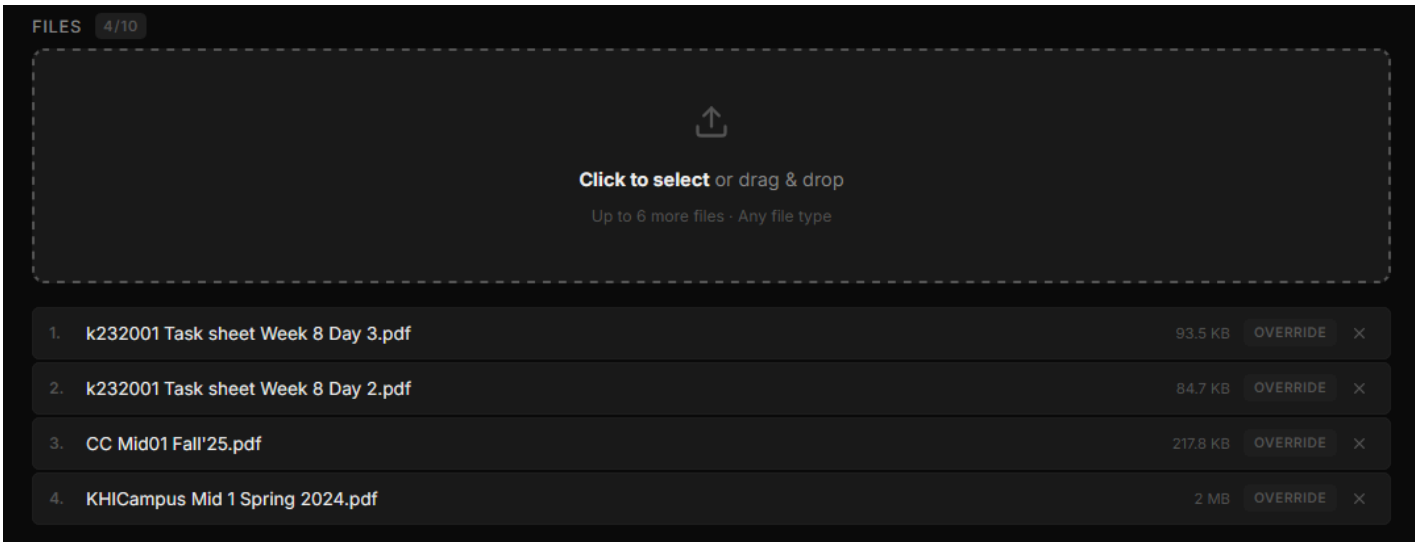
1. Sign in to your account (see Task 1). Contributions require a signed-in account.
2. Click Contribute in the navigation bar, or navigate directly to dasti.vercel.app/contribute.



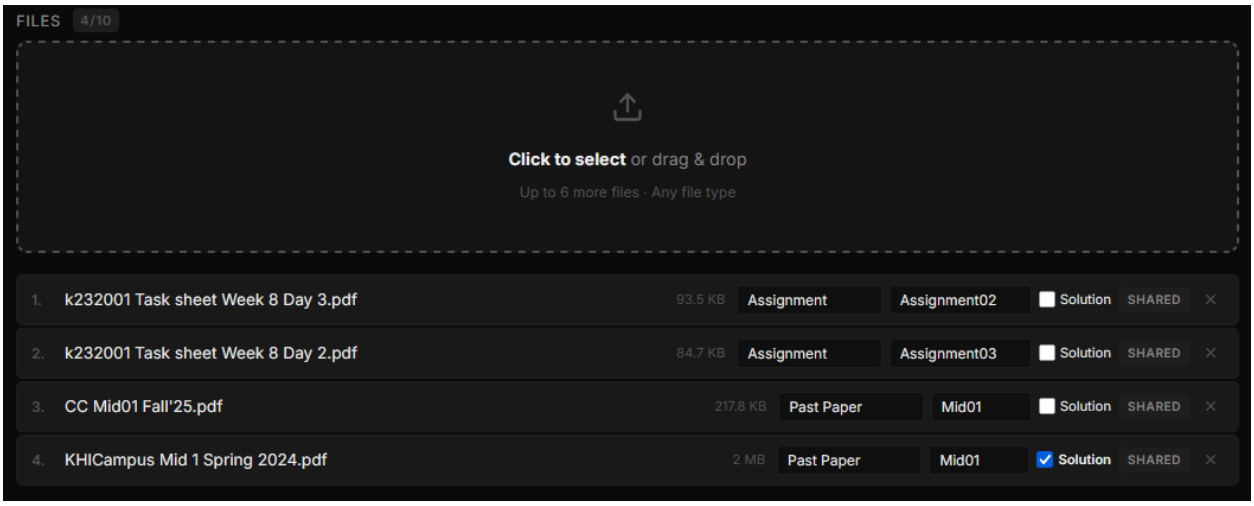
3. In the course search field, type the name, code, or shorthand of the course you are contributing to (for example, OOP, Object Oriented Programming, or CS2001). A dropdown of matching courses will appear.



4. Click the correct course from the dropdown. The Semester field will be filled in automatically.
5. Locate the file drop zone. Drag and drop your file(s) onto it, or click the zone to open a file browser. You may submit up to 10 files in a single submission.

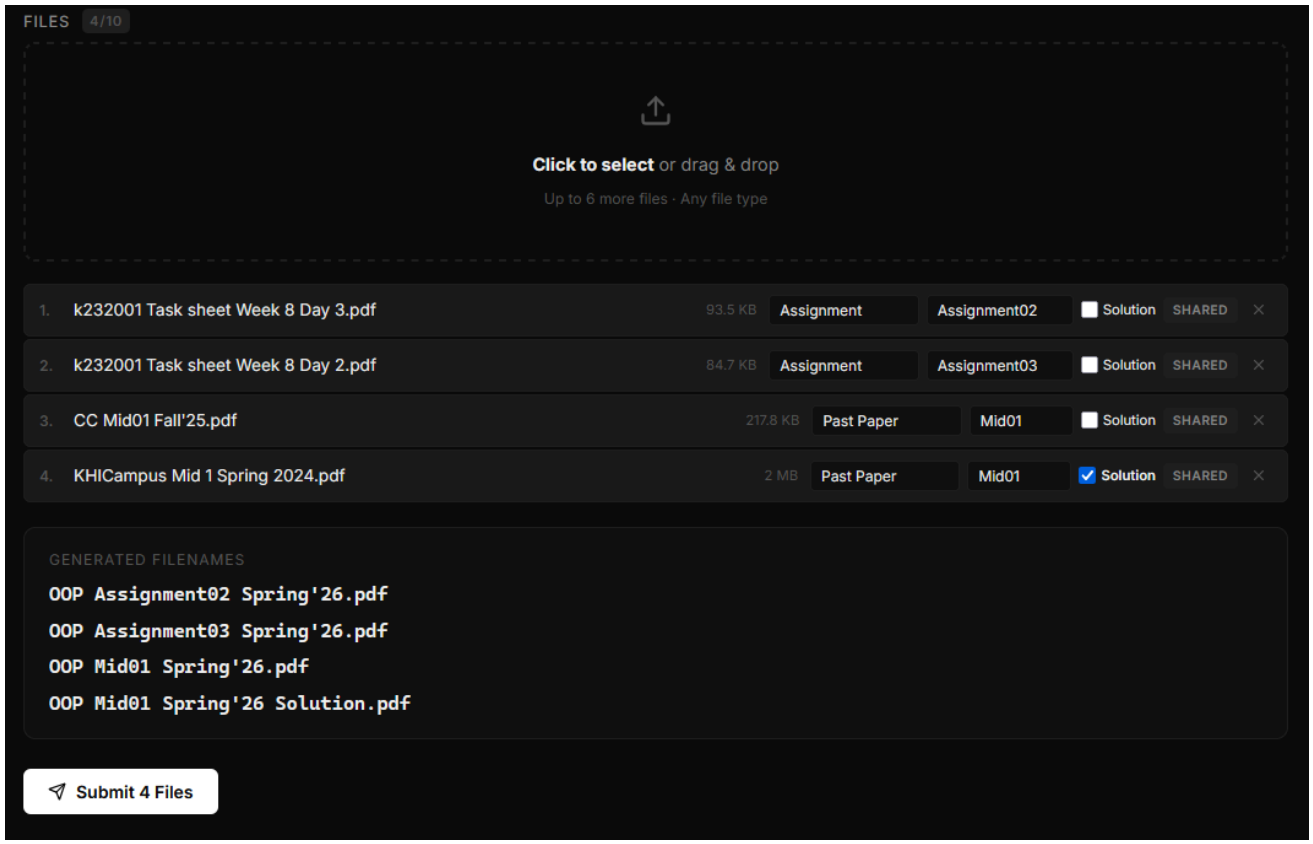


6. For each uploaded file, a row of metadata fields will appear (when override is selected). Fill in the following for each file: Material Type (Past Paper, Notes, Assignment, Quiz, Lab, Slides, or Other); Variant for Past Papers (Mid 1, Mid 2, or Final); and Semester/Year the material is from (for example, Fall 2024).

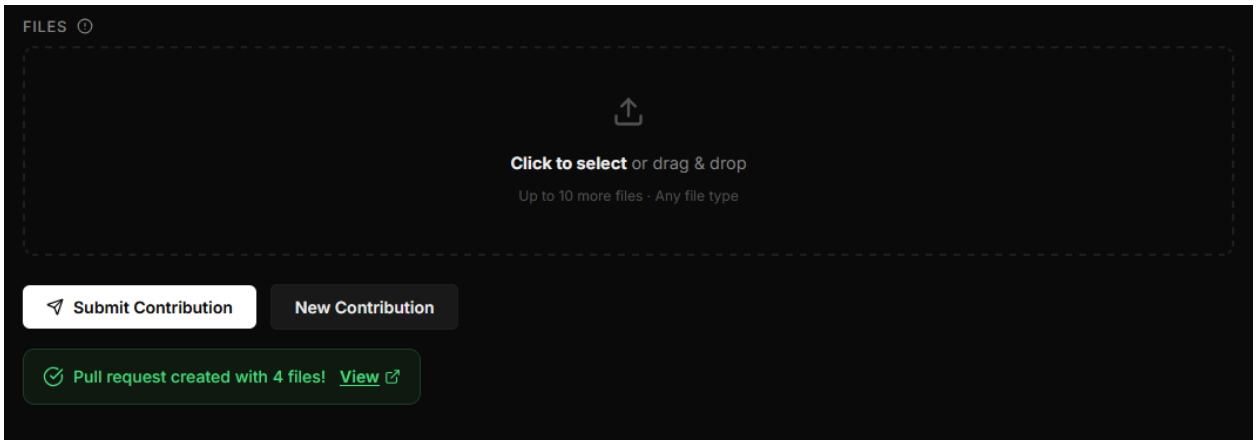


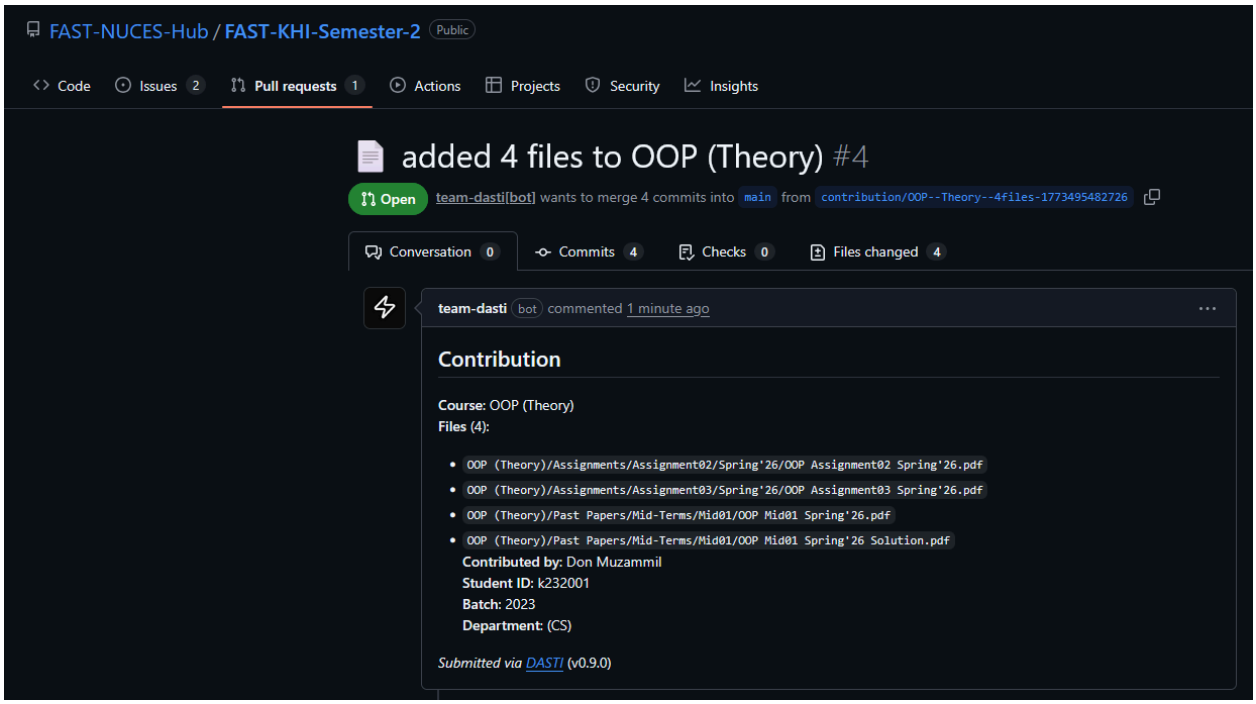
7. As you fill in the metadata, a filename preview will display the standardized name that will be used when committing the file (for example, OOP Mid01 Spring'26.pdf). Confirm this looks correct.

8. Once all files have their metadata filled in, click the Submit button.



9. Dasti will upload your files to a temporary staging area, then automatically create a branch on the semester repository, commit your files, and open a pull request. A success message will appear with a link to the pull request so you can track its status.





10. You will receive a submission confirmation email at your Google account address.
11. Once a repository maintainer reviews and merges your pull request, your contribution will be marked as approved and you will receive an approval email. Your leaderboard count will be updated.

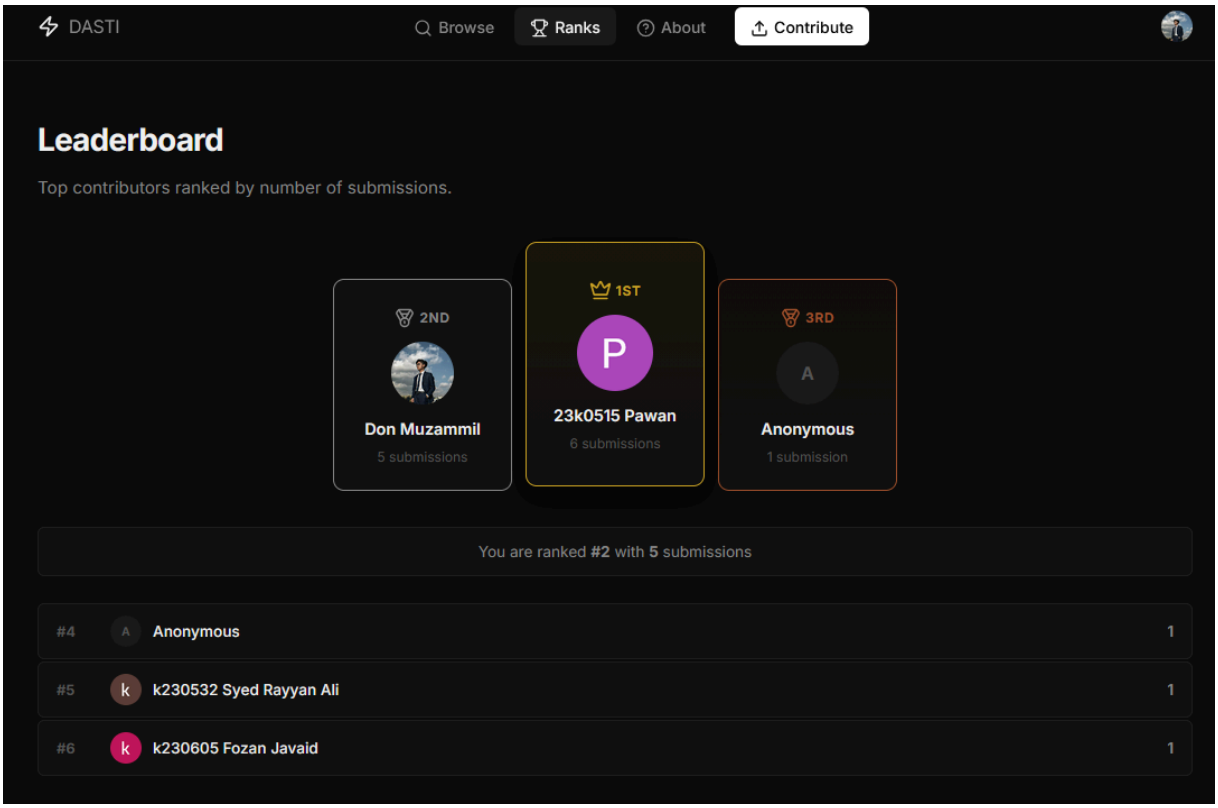
Important Notes:

- Accepted file types include PDF, DOCX, PPTX, XLSX, PNG, JPG, and common document formats. Executable files and other unsafe types are rejected.
- The platform allows 5 submissions per IP address per hour to prevent abuse.
- Duplicate files (identical content to another file in the same submission) are detected and rejected automatically.

Task 5: Tracking Your Contributions and the Leaderboard

Dasti keeps a record of your contributions and displays them on a public leaderboard.

1. Click Leaderboard in the navigation bar, or navigate to dasti.vercel.app/leaderboard.
2. The leaderboard lists contributors ranked by total number of contributions. Each entry shows the display name and contribution count.

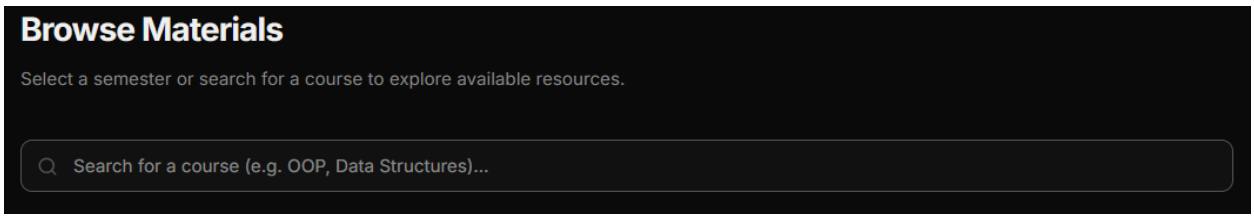


3. To view your own contributions specifically, navigate to dasti.vercel.app/profile while signed in. The profile page shows your personal contribution history including course, semester, file count, and status (Pending or Approved).
4. A contribution shows Pending until a repository maintainer merges the associated pull request. Once merged, the status changes to Approved and your leaderboard ranking is updated.

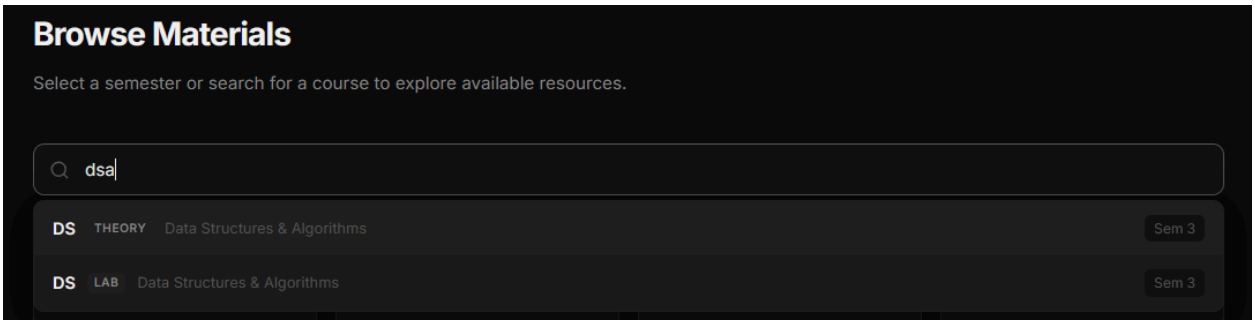
Task 6: Searching for a Specific Course

The Quick Search feature lets you jump directly to any course without navigating through the semester grid.

1. On the Browse page (dasti.vercel.app), locate the search bar at the top of the page.



2. Type a course name, code, or shorthand into the search bar (for example, DSA, Data Structures, or CS2002). Results appear as you type. The search is case-insensitive and works with partial input.



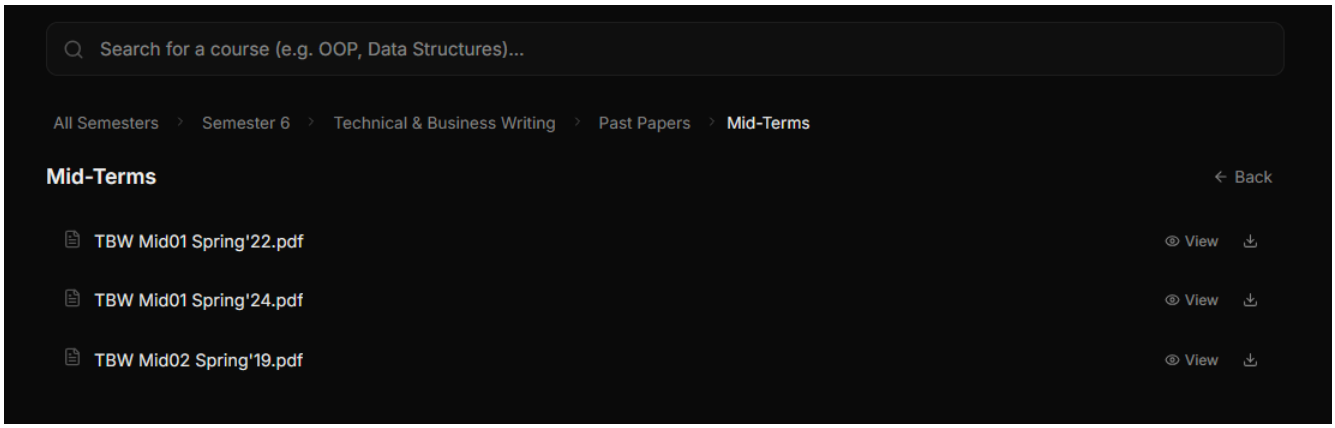
3. Click the course you are looking for. The page will navigate directly to that course's folder in the semester repository, bypassing the semester grid.
4. To clear the search and return to the full semester grid, delete the text in the search bar or click elsewhere on the page.

Note: *The same quick search is available on the Contribute page to help you find your course when submitting materials.*

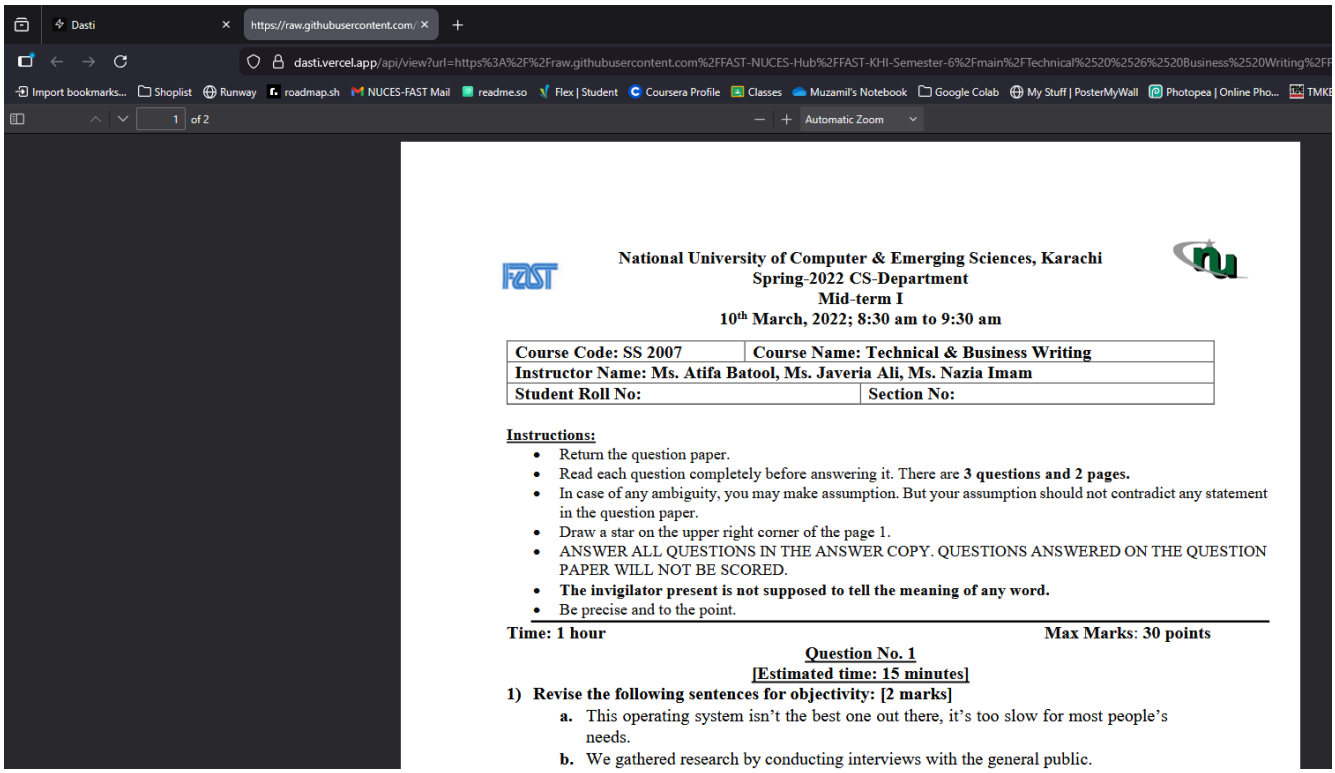
Task 7: Viewing and Downloading Files

Once you have navigated to a course folder using the Browse page, you can open or download individual files.

1. Navigate to the desired course folder (see Task 3 for navigation instructions).
2. Locate the file you want in the file list. Each file entry shows its name and two action buttons: View and Download.



3. To view a file in your browser without downloading it, click the View button. The file will open in a new browser tab. PDF files open directly; Microsoft Office files (DOCX, PPTX, XLSX) open in Google Docs Viewer automatically.



National University of Computer & Emerging Sciences, Karachi
Spring-2022 CS-Department
Mid-term I
10th March, 2022; 8:30 am to 9:30 am

Course Code: SS 2007	**Course Name: Technical & Business Writing**
Instructor Name: Ms. Atifa Batool, Ms. Javeria Ali, Ms. Nazia Imam	
Student Roll No:	**Section No:**

Instructions:

- Return the question paper.
- Read each question completely before answering it. There are **3 questions and 2 pages.**
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.
- Draw a star on the upper right corner of the page 1.
- ANSWER ALL QUESTIONS IN THE ANSWER COPY. QUESTIONS ANSWERED ON THE QUESTION PAPER WILL NOT BE SCORED.
- **The invigilator present is not supposed to tell the meaning of any word.**
- Be precise and to the point.

Time: 1 hour **Max Marks: 30 points**

Question No. 1
[Estimated time: 15 minutes]

1) Revise the following sentences for objectivity: [2 marks]

a. This operating system isn't the best one out there, it's too slow for most people's needs.

b. We gathered research by conducting interviews with the general public.

4. To download a file to your device, click the Download button. Your browser will begin downloading the file immediately with its standardized repository name (for example, DSA Final Spring'24.pdf).
5. If a file fails to open or download, try refreshing the page and attempting again. If the problem persists, refer to the Troubleshooting section.

Section 4: Troubleshooting

Problem 1: The Google Sign-In popup does not appear

Possible Cause: Your browser may be blocking popups from dasti.vercel.app.

Solution: Check your browser's address bar for a blocked popup notification (usually a small icon on the right side). Click it and select "Always allow popups from this site," then try again. If using an ad blocker extension, temporarily disable it for this site. If the issue continues, try a different browser such as Chrome or Firefox.

Problem 2: Submission rejected with "Rate limit exceeded"

Possible Cause: You or someone sharing your network connection has submitted more than 5 files within the past hour. Dasti enforces a limit of 5 submissions per IP address per hour to prevent misuse.

Solution: Wait until the hour window has passed and try again. If you are on a shared university network, try connecting through a different network such as your mobile data and resubmitting.

Problem 3: File rejected with an unsupported file type error

Possible Cause: The file you attempted to upload is not in one of the accepted formats. Dasti only accepts standard academic document types such as PDF, DOCX, PPTX, XLSX, and common image formats. Compressed archives, executable files, and other non-document types are not accepted.

Solution: Convert your file to PDF before uploading. Most word processors (Microsoft Word, Google Docs, LibreOffice) allow you to export via File > Save As or File > Export. Re-upload the converted PDF.

Problem 4: Contribution shows "Pending" for a long time

Possible Cause: Pull requests are reviewed and merged manually by the repository maintainer. There is no automated merging. This may take anywhere from a few hours to several days, particularly around exam season when submission volumes are higher.

Solution: No action is needed on your part. Your contribution is safely stored and the pull request is open. You can click the pull request link from your profile page to monitor its status directly on GitHub. If it has been more than a week with no activity, contact the platform maintainer through the About page.

Problem 5: Clicking "View" opens a blank page or shows an error

Possible Cause: The file may be temporarily unavailable due to a GitHub API issue, or the file format may not be supported by the in-browser viewer.

Solution: Wait a few minutes and try again. If the file is a Microsoft Office format and Google Docs Viewer is showing an error, use the Download button instead and open the file locally. If the issue persists across multiple files, there may be a temporary outage with GitHub's content API.

Section 5: Glossary

Branch

An independent line of development within a GitHub repository. When Dasti submits your files, it creates a new branch on the semester repository specifically for your contribution, then opens a pull request to merge that branch into the main repository. This keeps the main repository clean until a maintainer approves the changes.

Filename Convention

A standardized format used by Dasti when naming files committed to the semester repositories. The format follows the pattern “ShortName Type Semester.pdf”. For example, a mid-term past paper for Object Oriented Programming from Fall 2024 would be named “OOP Mid01 Fall’24.pdf”, ensuring consistency across all contributions.

GitHub App

A type of GitHub integration that can be granted specific permissions on repositories. Dasti uses a GitHub App to interact with the semester repositories. The app has only the minimum permissions needed - it can create branches and open pull requests, but it cannot merge or delete content.

Leaderboard

A ranked list of Dasti users sorted by the number of approved contributions they have made to the semester repositories. The leaderboard is publicly visible without signing in.

OAuth (Open Authorization)

A standard protocol that allows a user to grant a third-party application access to their account information without sharing their password. Dasti uses Google OAuth so that you can sign in with your Google account without creating a separate username and password for Dasti.

Pull Request (PR)

A formal proposal to add changes to a repository on GitHub. When you submit a file on Dasti, the platform automatically opens a pull request on the appropriate semester repository. A repository maintainer then reviews the files and, if appropriate, merges the pull request - which is the act of officially adding your files to the repository.

PWA (Progressive Web App)

A type of web application that can be installed on a device and used similarly to a native app. Dasti is configured as a PWA, meaning you can add it to your phone or desktop home screen for quick access.

Rate Limiting

A security measure that restricts the number of requests a single user or IP address can make within a given time window. Dasti limits file submissions to 5 per IP address per hour to prevent automated abuse.

Repository (Repo)

A storage location on GitHub where files and their history are kept. FAST-KHI maintains separate repositories for each semester (for example, FAST-KHI-Semester-2), and Dasti reads from and writes to these repositories on your behalf.

Semester Repository

One of the GitHub repositories maintained by FAST-NUCES-Hub that contains academic materials for a specific semester. There are six active semester repositories, each corresponding to one semester of the FAST-KHI curriculum.

Supabase

A backend platform used by Dasti to store user account information, contribution records, and temporarily stage files before they are committed to GitHub. Users do not interact with Supabase directly; it operates entirely behind the scenes.

Webhook

An automated message sent from one system to another when a specific event occurs. Dasti uses a GitHub webhook so that when a pull request is merged by a repository maintainer, GitHub automatically notifies Dasti's server, which then updates the contribution status to Approved and sends an approval email to the contributor.

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Duplicate file detection	Task 4 (Important Notes)
Email notifications	Task 1 (Step 5), Task 4 (Steps 11–12)
File drop zone	Task 4 (Step 6)
File types (accepted)	Task 4 (Important Notes), Troubleshooting Problem 3
Filename convention	Task 4 (Step 8), Glossary
Folder selection (Theory/Lab)	Task 3 (Step 3), Task 4 (Step 5)
GitHub App	Glossary
Google OAuth / Sign-In	Task 1, Glossary
Guest access	Task 1 (Note)
Leaderboard	Task 5, Glossary
Material type (metadata)	Task 4 (Step 7)
Merge (pull request)	Task 4 (Step 12), Troubleshooting Problem 4
Metadata fields	Task 4 (Step 7)
PDF conversion	Troubleshooting Problem 3
Pending status	Task 5 (Step 4), Troubleshooting Problem 4
Profile page	Task 2, Task 5
Pull request	Task 4 (Steps 10–12), Glossary
PWA	Glossary

Rate limiting	Task 4 (Important Notes), Troubleshooting Problem 2, Glossary
Repository	Glossary
Saving profile	Task 2 (Step 6)
Search bar (Quick Search)	Task 6
Semester grid	Task 3 (Step 1)
Semester repository	Glossary
Sign in	Task 1
Submission confirmation	Task 4 (Steps 10–11)
Supabase	Glossary
Viewing files in browser	Task 7 (Step 3)
Webhook	Glossary

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